Gabriel Claret

PhD student in applied mathematics

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Research interests

(i) Analysis of Partial Differential Equations on Sobolev extension domains

(ii) Potential theory

(iii) Inverse problems

(iv) Domain convergence

Training and diplomas

Since 2023 **PhD student in applied mathematics**, *Fédération de Mathématiques (CNRS FR-3487) and MICS laboratory, CentraleSupélec, Université Paris-Saclay*, Gif-sur-Yvette, France
Title: Imaging and inverse scattering problems with non-Lipschitz absorbing boundaries.
Advised by Anna Rozanova-Pierrat.

2022 – 2023 **Master's degree in Mathematics of Randomness**, *Institut de Mathématiques d'Orsay, Université Paris-Saclay*, Orsay, France
'Probability and Statistics' track.

2020 – 2023 **Engineering degree**, *CentraleSupélec, Université Paris-Saclay*, Gif-sur-Yvette, France
Major: 'Mathematics and Data Science'; Minor: 'Mathematical Modeling and Financial Mathematics'; Mathematics track.
'Research careers' stream, 'Parcours recherche' label.

2018 – 2020 **Classes Préparatoires aux Grandes Écoles**, *Lycée Saint-Louis*, Paris, France
A two-year intensive curriculum to prepare for the French graduate schools competitive entrance exams.
Mathematics and Physics track.

2018 **Scientific *Baccalauréat***, *Lycée Léonard de Vinci*, Levallois-Perret, France
Mathematics, Physics and Biology-Geology track, Mathematics specialty.
English European section – *a track which focuses on the English language and cultures*.

Internships and other research experiences

2023 **End-of-studies internship**, *Laboratoire MICS, CentraleSupélec, Université Paris-Saclay*, Gif-sur-Yvette, France
Title: Functional, spectral and probabilistic analysis on Sobolev extension domains.
Supervised by Anna Rozanova-Pierrat (6 months).

2020 – 2023 **Parcours recherche**, *CentraleSupélec, Université Paris-Saclay*, Gif-sur-Yvette, France
A three-year research project carried out in parallel of the engineering curriculum in CentraleSupélec.
Title: Inverse and imaging problems for object with irregular or fractal boundaries.
Supervised by Anna Rozanova-Pierrat.

2022 **Research internship**, *University of Bielefeld*, Bielefeld, Allemagne
Title: Calderón's problem on non-Lipschitz and fractal domains, and convergence of the Poincaré-Steklov operators.
Supervised by Michael Hinz (6 months).

Scientific productions

Accepted papers

In press **Layer potential operators for transmission problems on extension domains**, *with Michael Hinz, Alexander Teplyaev and Anna Rozanova-Pierrat*, Journal des Mathématiques Pures et Appliquées
Keywords: fractal boundary, layer potential operators, Poincaré-Steklov (Dirichlet-to-Neumann) operator, Neumann-Poincaré operator, Calderón projectors, elliptic transmission problem, inverse problems.
Open access: [HAL]

2025 **Existence of optimal shapes for heat diffusions across irregular interfaces**, *with Anna Rozanova-Pierrat*, in Fractal Geometry in Pure and Applied Mathematics, Contemporary Mathematics (CONM) book series, American Mathematical Society.
Keywords: heat equation, shape optimisation, trace operators, extension domains, fractals, Mosco convergence.
Open access: [HAL]

Preprints

2026 **Helmholtz transmission problem and intrinsic impedance scattering problem on extension domains**, submitted
Keywords: Helmholtz equation, acoustic scattering, transmission problem, Robin boundary condition, layer potential operators.
Open access: [HAL]

2025 **Poincaré-Steklov operator and Calderón's problem on extension domains**, *with Michael Hinz and Anna Rozanova-Pierrat*, preprint
Open access: [HAL]

2024 **Convergence of layer potentials and Riemann-Hilbert problem on extension domains**, *with Anna Rozanova-Pierrat and Alexander Teplyaev*, submitted
Keywords: Sobolev extension domains, layer potential operators, Neumann-Poincaré operators, Neumann series, Cauchy integral, Hilbert transform, dyadic approximation, domain convergence.
Open access: [HAL]

Other

2023 **Functional, spectral and probabilistic analysis on Sobolev extension domains**
Master's thesis, advised by Anna Rozanova-Pierrat
Keywords: Sobolev extension domains, traces and weak normal derivatives, elliptic transmission problem, layer potential operators, Poincaré-Steklov (Dirichlet-to-Neumann) operator, Neumann-Poincaré operator, imaging problem, Hilbert transform and Cauchy integral, Dirichlet forms and Markov chains.
Open access: [DUMAS]

Responsibilities

Apr. 2025 **Co-organiser of the conference 'Kinetic equations and turbulence'**, *in honour of Claude Bardos's 85th birthday*, Gif-sur-Yvette, France
Conference organised in CentraleSupélec, satellite workshop in the Collège de France.
Co-organised with: Anna Rozanova-Pierrat, François Glose, Daniel Bouche and Norbert Mauser.
Website: https://bardos-85.sciencesconf.org/?lang=en

Scientific presentations

Talks

Mar. 2026 **Congress CJC-MA 2026**, *École Nationale des Ponts et Chaussées*, Champs-sur-Marne, France
Talk entitled 'Optimisation for the intrinsic impedance scattering problem on extension domains'.

Jun. 2025 **Cornell Conference on Analysis, Probability, and Mathematical Physics on Fractals**, *Cornell University*, Ithaca, USA
Keynote talk entitled 'The Riemann-Hilbert problem on extension domains'.

May 2025 **Visit by the CNRS INSMI of the Fédération de Mathématiques**, *CentraleSupélec, Université Paris-Saclay*, Gif-sur-Yvette, France
Talk entitled 'Le problème de Riemann-Hilbert sur domaines d'extension'.

Dec. 2024 **MICS laboratory winter party**, *CentraleSupélec, Université Paris-Saclay*, Gif-sur-Yvette, France
Talk entitled 'Approximating problems involving irregular and fractal shapes: a travel from space to space'.

Jun. 2024 **MICS laboratory summer party**, *CentraleSupélec, Université Paris-Saclay*, Gif-sur-Yvette, France
Talk entitled 'Irregular shapes in mathematical problems: why and how'.

Mar. 2024 **Analysis on fractals and networks, and applications**, *CIRM*, Marseille, France
Talk entitled 'Poincaré-Steklov and Neumann-Poincaré operators for extension domains'.

Jan. 2024 **Rencontres MathTech 2024**, *IHES*, Bures-sur-Yvette, France
Pitch entitled 'Dealing with the irregular in imaging problems'.

Nov. 2023 **Seminar of the Mathematics Federation**, *CentraleSupélec, Université Paris-Saclay*, Gif-sur-Yvette, France
Seminar entitled 'Layer potentials and imaging on irregular domains'.

Poster

Sep. 2023 **Congress CJC-MA 2023**, *CentraleSupélec, Université Paris-Saclay*, Gif-sur-Yvette, France
Poster entitled 'Dealing with the irregular in inverse and imaging problems'.
Open access: [HAL]

Teachings

2023-2026 **Teachings in CentraleSupélec**, *CentraleSupélec, Université Paris-Saclay*, Gif-sur-Yvette, France

In France, the undergraduate programme is divided into three years of 'licence' (L1, L2, L3), followed by two years of 'master' (M1, M2).

Type	Level	Description	Total	Years	Language
Tutorial	L3	Convergence, Integration and Probabilities	45h	23-26	FR
		Partial Differential Equations	45h	23-26	FR
Project	M1	Control of the acoustic and electromagnetic pollution	81h	23-26	FR
Mentoring*	L1	Mathematics for the Bachelor of Global Engineering	40h	23-25	EN
Tutorial	L1	Mathematics for the Bachelor of Global Engineering	8h	25-26	EN
Lecture + Tutorial	M2	Weak convergence and compact operators	3h	23-24	FR
Tutorial	M1	Elliptic problems on irregular domains	3h	23-24	FR

*Mentoring sessions and coordination of the mentoring team.

Reviews

2025-2026 **Integral Transforms and Special Functions**, Taylor & Francis

2025 **Gazette de la Société Mathématique de France**

Participation to scientific events

Mar. 2026 **Congress CJC-MA 2026**, *École Nationale des Ponts et Chaussées*, Champs-sur-Marne, France

Jan. 2026 **'Jeunes Edpistes en France 2026' days**, *Université Paris Dauphine-PSL*, Paris, France

Apr. 2025 **Conference 'Kinetic equations an turbulence'**, *CentraleSupélec and Collège de France*, Gif-sur-Yvette and Paris, France

Jan. 2025 **'Jeunes Edpistes en France 2025' days**, *Université Côte d'Azur*, Nice, France

Mar. 2024 **Conference 'Fractals, quantum graphs and applications in pure and applied sciences'**, *Politecnico di Milano*, Milan, Italie

Mar. 2024 **Conference 'Analysis on fractals and networks, and applications'**, *CIRM*, Marseille, France

Jan. 2024 **Rencontres MathTech 2024**, *IHES*, Bures-sur-Yvette, France

Dec. 2023 **SALVE winter school**, *École Normale Supérieure*, Paris, France

Sep. 2023 **Congress CJC-MA 2023**, *CentraleSupélec, Université Paris-Saclay*, Gif-sur-Yvette, France

Other skills

Languages

French	Native	
English	Bilingual	*Level C2 (8,5/9 at the IELTS in 2021)*
Spanish	Fluent	
Italian	Notions	
German	Notions	

Computer skills

LaTeX, Suite Office, Python